

A five-layer validation framework for computational synthetic-lethality discovery: from an MBD4-loss anchor case to a cross-platform-replicated pan-cancer portfolio

Fahad Kiani*
CrisPRO.org
Yusuf Mohammed
Rutgers University
Ali Khan
Rutgers University

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Abstract

Computational synthetic-lethality (SL) discovery from cancer-dependency screens can nominate genotype-selective drug targets at scale, but nominations are frequently confounded, non-reproducible, or clinically uninterpretable. We define and apply a five-layer validation framework that separates a raw dependency signal from an actionable, evidence-graded SL hypothesis: (L1) recovery of a curated known-SL truth set; (L2) genome-wide differential-dependency discovery; (L3) orthogonal confound stress-testing; (L4) cross-screen integration-robustness plus cross-platform independent replication; and (L5) a clinical evidence ladder with a two-dimensional trust-tier assignment and explicit actionability. We first apply the framework to a fully worked anchor case, MBD4 loss-of-function (LOF), a base-excision-repair (BER) glycosylase defect. In DepMap 24Q2, MBD4-LOF cell lines (n=21; 15 TP53 co-mutated, 6 MSI-H) show the gold-standard cytidine-analog SL interaction (isogenic rescue in the literature) and a replication-stress-driven ATR-checkpoint dependency (ceralasertib LN_IC50 Delta=-0.73, one-sided p=0.022, Cohen's d=-0.50, n=14 LOF vs 922 WT; robust to four confound stress tests), while a transcriptional PARP1-upregulation model for PARP-inhibitor vulnerability is falsified (no elevation; one-sided Mann-Whitney p=0.27, two-sided p=0.53; n=19 LOF) and therefore retained here only as a cautionary vignette that motivates orthogonal validation. We then generalize to a pan-cancer portfolio of 58 driver->partner SL

*fahad@crispro.ai

axes. Across an orthogonal-technology benchmark (Sanger Project Score CERES, KY1.0/1.1 library, versus Broad Avana+Chronos discovery), 51 of 58 axes were testable and 41 replicated at $FDR < 0.05$ with strong effect-size concordance (Spearman $\rho = 0.791$, $p = 5.1 \times 10^{-12}$; Pearson $r = 0.813$, $p = 4.5 \times 10^{-13}$; $n = 51$); a separate within-platform integration-robustness check gave $\rho = 0.973$. The evidence ladder graded 4 axes E1 (approved drug in matched context), 8 E2, 6 E3, 10 E4, and 30 E5 (computational-only), and the trust-tier scheme identified 5 axes that are both clinically anchored and replicated. We state the honest ceiling explicitly: CRISPR- derived SL cannot be clinically validated in this study, panel overlap means the L4 replication is cross-platform rather than out-of-cohort, and no axis is labeled clinically proven. The framework is a reproducibility and interpretability contract for turning dependency signals into graded, auditable SL hypotheses. Research Use Only.

Introduction

Synthetic lethality (SL) — a genetic interaction in which the co-occurrence of two perturbations is lethal while either alone is tolerated — is one of the few mechanistic levers that converts a tumor’s loss-of-function genotype into a selective therapeutic vulnerability (O’Neil et al. 2017; Huang et al. 2020). The clinical validity of the paradigm is established by PARP inhibition in *BRCA1/2*-mutant cancers (Lord and Ashworth 2017; Farmer et al. 2005; Bryant et al. 2005), and genome-scale CRISPR-Cas9 loss-of-function screens across large cancer cell-line panels (DepMap; Project Score) now make it possible to nominate genotype-selective dependencies systematically (Tsherniak et al. 2017; Behan et al. 2019; Meyers et al. 2017).

The difficulty is not generating nominations — it is trusting them. Differential-dependency signals are routinely confounded by co-occurring features such as microsatellite instability (MSI), *TP53* status, copy-number, and lineage composition; effect sizes drift between screening pipelines; and a statistically significant dependency is not, by itself, a druggable or clinically interpretable hypothesis. Reviews of functional-genomics SL screens repeatedly stress reproducibility, confounding, and context-specificity as the dominant failure modes (O’Neil et al. 2017; Ryan et al. 2018). A nomination pipeline that does not separate “a signal survived a t-test” from “an orthogonally replicated, confound-controlled, evidence-graded, actionable hypothesis” will over-promise.

Here we make that separation explicit. We define a **five-layer validation framework** and apply it end to end: first to a single, fully worked anchor case — MBD4 loss-of-function — where we can show each layer against concrete receipts; then to a pan-cancer portfolio of 58 driver->partner SL axes, where the same layers are applied at scale. Throughout, we foreground the honest ceiling of what dependency data can and cannot prove, and we tag every quantitative claim to a machine-readable receipt.

This manuscript is deliberately scoped to the **synthetic-lethality** program. The MBD4 anchor has a non-SL immunotherapy dimension (a hypermutator/immune-checkpoint hypothesis) that is treated elsewhere; it is out of scope here. We use MBD4 strictly as the SL worked example, including the *falsified* PARP1 mechanism, because a validation framework is only credible if it also reports what it rules out.

The five-layer validation framework

We define the framework verbatim so the term is unambiguous. Each layer answers a distinct question and has an explicit pass/fail or grading criterion; a nomination advances only as far as the evidence supports.

- **Layer 1 — Known-SL truth-set recovery (calibration).** Can the pipeline recover curated, independently established SL interactions from the same dependency data? This calibrates sensitivity and exposes modality blind spots (e.g., drug-trapping mechanisms that gene knockout cannot model).
- **Layer 2 — Discovery by differential dependency.** Genome-wide, genotype-stratified comparison of dependency (Chronos/CERES gene-effect) between driver-altered and wild-type lines, ranked by effect size and significance, to nominate candidate driver->partner axes.
- **Layer 3 — Confound stress-testing.** Each promoted axis is challenged against the confounders that dominate SL false positives: MSI status, *TP53* co-mutation, lineage composition, and single-line leverage (leave-one-out). A signal that inverts, evaporates, or is carried by one line or one lineage does not advance.
- **Layer 4 — Cross-screen integration-robustness and cross-platform independent replication.** Two distinct checks: (a) *integration-robustness* — does the signal survive the screen-integration pipeline (per-screen effect vs integrated model on the same platform)? and (b) *cross-platform replication* — does it reproduce in an orthogonal library + wet-lab + analysis pipeline (Sanger Project Score CERES vs Broad Avana+Chronos)? We report these separately and never conflate them.
- **Layer 5 — Clinical evidence ladder, trust tiers, and actionability.** Each axis is graded on a clinical-evidence ladder (E1 approved drug in matched context -> E5 computational-only nomination), assigned a two-dimensional trust tier that crosses clinical anchoring with replication status, and annotated for druggability/actionability against curated drug and pharmacogenomic resources.

Layers 1-3 correspond to the confound-validated discovery layer established previously for the anchor case; Layers 4-5 are the external-validation and trust-scoring layer added for the pan-cancer portfolio. The framework is a contract:

it fixes, in advance, what has to be true for a nomination to be reported at each level of confidence.

Anchor case: MBD4 loss-of-function

MBD4 (methyl-CpG-binding domain protein 4) is a BER glycosylase that excises thymine from G:T mismatches generated by spontaneous deamination of 5-methylcytosine at CpG sites; its loss produces a CpG>TpG hypermutator phenotype and BER deficiency (Hendrich et al. 1999; Bader et al. 1999; Krokan and Bjoras 2013; Wallace et al. 2012). We use MBD4-LOF as the anchor because it exercises every layer of the framework with concrete receipts, and because it includes a genuine falsification.

Cohort (receipts). In DepMap 24Q2, loss-of-function status was called exclusively from the LikelyLoF annotation (missense/passenger variants purged at the classification gate), yielding n=21 MBD4-LOF lines, of which 15 are *TP53* co-mutated and 6 are MSI-H. Drug-screen intersections with GDSC2 reduce per-drug denominators (ceralasertib n=14; adavosertib n=15).

Layer 1 (calibration) — cytidine-analog SL is the gold standard. The MBD4-cytidine-analog interaction is an isogenic, rescue-confirmed SL: MBD4-knockout cells are hypersensitive to gemcitabine and cytarabine, with sensitivity reverting on MBD4 re-expression (Chabot and colleagues 2022). This is the strongest evidence class and sets the calibration bar against which downstream findings are judged.

Layer 2-3 (discovery + confounds) — a replication-stress ATR dependency. Genotype-stratified GDSC2 analysis shows MBD4-LOF lines are significantly more sensitive to the ATR inhibitor ceralasertib (AZD6738) than wild-type: LN_IC50 Delta=-0.73 (one-sided p=0.022, Cohen's d=-0.50; n=14 LOF vs 922 WT; two-sided p=0.044), directionally concordant across all three GDSC2 potency metrics (AUC Delta=-0.06, one-sided p=0.013, d=-0.55; Z-score Delta=-0.50, one-sided p=0.022, d=-0.50). The WEE1 inhibitor adavosertib shows the same direction sub-significantly (LN_IC50 Delta=-0.51, one-sided p=0.074). The ceralasertib signal survives four orthogonal confound stress tests: it *strengthens* after removing MSI-H lines (Delta=-0.91, one-sided p=0.015), remains large after isolating the MBD4 contribution within a *TP53*-mutant background (Delta=-1.07, one-sided p=0.003, d=-0.74), retains one-sided p<0.05 in 14/14 leave-one-out iterations, and is preserved outside the dominant lineage (non-bowel Delta=-0.87, one-sided p=0.025). All anchor drug-response tests are one-sided (directional SL hypothesis: LOF more sensitive). This is the Layer-3 pattern a real driver->partner axis should produce.

A falsification, retained as a cautionary vignette. It has been proposed that BER deficiency drives compensatory PARP1 upregulation, creating a PARP-trapping vulnerability. In DepMap 24Q2 expression, MBD4-LOF lines show **no** PARP1 elevation (n=19 LOF with expression, out of 21 MBD4-

LikelyLoF models; median 6.77 vs 6.64 in non-LOF lines): the hypothesized *upregulation* direction is not supported (one-sided Mann-Whitney $p=0.27$; two-sided $p=0.53$), and the proposed RNF144A degradation bridge is likewise absent (one-sided $p=0.28$; two-sided $p=0.56$). The specific transcriptional-upregulation mechanism is therefore falsified for MBD4. Separately, PARP1 expression is a strong pan-cancer predictor of PARP-inhibitor sensitivity (Spearman $\rho=-0.42$, $p=5.2 \times 10^{-22}$, $n=488$; GDSC2), so the marker is real but is *not* switched on by MBD4 status. We keep this vignette in the SL manuscript precisely because it demonstrates Layer-3 discipline: an attractive, mechanistically plausible SL hypothesis that the data do not support is reported as falsified, not quietly dropped. PARP inhibition is not advanced as an MBD4 SL axis on this basis.

The anchor therefore yields two supported SL levers (cytidine analogs, gold standard; ATR inhibition, confound-validated) converging on the replication fork, and one explicitly ruled-out mechanism. It is the template for how a single axis should look after Layers 1-3.

Generalization: a pan-cancer 58-axis SL portfolio

We applied the same framework at scale to 58 driver->partner SL axes surviving discovery and confound stress-testing. Discovery used the DepMap 24Q4 integrated Chronos matrix (CRISPRGeneEffect, 1178 lines x 17,916 genes); mutation calls from DepMap 24Q4 OmicsSomaticMutations; and drug/pharmacogenomic annotation from the Broad Drug Repurposing Hub and ClinPGx.

Layer 1 at scale: known-SL recovery and modality limits

Against a curated truth set of 27 known SL pairs (4 approved, 13 pivotal-trial, 10 preclinical-validated), the CRISPR-visible recall was $10/22 = 0.455$, with recovery concentrated in the higher-evidence tiers (approved 1/2; pivotal-trial 6/11; preclinical 3/9). PARP-modality pairs were recovered at only 1/5 and are excluded from primary recall *by design*, because gene knockout cannot model PARP trapping — the same modality blind spot the MBD4 vignette illustrates. All misses are mechanistically explained (modality, partner absent from the library, or copy-number axis). Precision is not meaningfully computable against a truth set dominated by novel positives; we therefore report recall and reclassification-correctness only.

Layer 4 at scale: integration-robustness vs cross-platform replication

We report two distinct L4 checks and keep them strictly separate.

(a) Within-platform integration-robustness. Comparing per-screen gene effect (DepMap 24Q4 ScreenGeneEffect, 1186 models) to the integrated Chronos model, 53/53 mutation-testable survivors preserve sign and significance, with effect-size Spearman $\rho=0.973$ ($p=4.8 \times 10^{-34}$), Pearson $r=0.99$. This confirms

that survivors are not artifacts of the integration pipeline. It is a **necessary robustness condition, not independent replication**: the per-screen matrix shares 1177/1186 lines with discovery (only 9 truly held out).

(b) Cross-platform independent replication. We replicated against Sanger Project Score CERES (Behan et al. 2019; Figshare 10.6084/m9.figshare.9116732; KY1.0/1.1 library (Tzelepis et al. 2016)) — an orthogonal guide library, wet-lab, and analysis pipeline versus Broad Avana+Chronos. Of 58 axes, 51 were testable; 47/51 were sign-concordant and **41 replicated at FDR<0.05**, with strong effect-size concordance: Spearman $\rho=0.791$ ($p=5.1\times 10^{-12}$), Pearson $r=0.813$ ($p=4.5\times 10^{-13}$), $n=51$ (Fig. 1). Positive-control axes behaved as expected (PTEN->PIK3CB, BRAF->MAPK1, BRAF->MAP2K1, STAG2->STAG1 all replicated; VHL->EPAS1 was insufficient_n). Non-testable axes were accounted for (3 insufficient n; 4 copy-number axes not testable in the CERES matrix).

Honesty note (critical). The Sanger CERES Figshare version is pre-filtered to lines with Broad copy-number data, so the panels overlap heavily (315/318 shared; 0 held-out lines were testable for mutation-stratified analysis). The $\rho=0.791$ result is therefore **cross-platform replication (orthogonal technology on shared lines), not out-of-cohort replication**. We deliberately report this weaker, honest number as the headline external-validation statistic rather than the within-platform $\rho=0.973$.

Layer 5 at scale: evidence ladder, trust tiers, actionability

Each axis was placed on a clinical-evidence ladder (Fig. 2): **E1** (approved drug in matched context) = 4 axes; **E2** (clinical-trial precedent) = 8; **E3** (pre-clinical in vivo) = 6; **E4** (preclinical in vitro / paralog-established) = 10; **E5** (computational-only nomination) = 30; total 58. Cross-tabulating grade against replication status shows that higher-evidence axes are also the ones that replicate cross-platform (e.g., all 6 E3 axes and 8/10 E4 axes replicated), whereas the 30 E5 axes are discovery- only nominations.

A two-dimensional trust tier crossed clinical anchoring with replication: **5 axes are both clinically anchored and replicated** (T1: BRAF->MAPK1, BRAF->MAP2K1, PTEN->PIK3CB, STK11->HDAC4, BRCA1->PARP1), 21 are anchored with precedent (T2), 22 are replicated without precedent (T3), 2 are precedent-only, and 8 are discovery-only (T4). Four of the five T1 axes clear the Layer-1 CRISPR discovery gate directly (FDR from 1.1×10^{-14} to 3.8×10^{-5}). The fifth, BRCA1->PARP1, does *not* clear that gate (Layer-1 FDR=0.25; internally tiered “Nominal”) because gene knockout cannot model PARP trapping — the same modality blind spot the MBD4 vignette illustrates — and it enters T1 only via its approved-drug clinical anchor plus cross-platform support; we flag it explicitly rather than let it read as a CRISPR-discovered hit. (This T1 set uses the primary two-dimensional trust scheme; an alternative one-dimensional clinical-anchor tiering that ignores replication status labels

12 axes and is reported in the supplementary receipts only.) Actionability annotation (MOA-consistency-filtered against the Drug Repurposing Hub and ClinPGx) found 13/58 axes with an MOA-consistent selective druggable agent at any phase, 9/58 at late phase (Phase 3+/approved), 17/58 with druggable or trial precedent, 23/58 with a ClinPGx-annotated partner, and 5/58 with an FDA/EMA label. Literature was formally searched for 12 axes; the remaining 46 are unexamined (which is not the same as “no precedent”).

Discussion

The contribution of this work is a **validation contract**, not a single dependency hit. By fixing five layers in advance — calibration, discovery, confound stress-testing, dual-mode replication, and clinical grading with trust tiers — the framework forces every SL nomination to declare exactly how far the evidence carries it, and makes the whole chain auditable against receipts.

The MBD4 anchor shows why each layer is load-bearing. Layer 1 (cytidine-analog gold standard) calibrates the pipeline. Layers 2-3 promote the ATR dependency only after it strengthens through an MSI purge, survives *TP53* stratification, and passes leave-one-out and lineage controls — the exact confounds that sink naive SL screens. And the PARP1 falsification (no upregulation; one-sided $p=0.27$) demonstrates the framework reporting a negative result rather than a convenient one. At scale, the same layers separate 5 clinically anchored, replicated axes from 30 discovery-only nominations, so downstream users can prioritize accordingly.

The most important result to read correctly is the L4 replication. We report **two** numbers because they answer two questions. The within-platform $\rho=0.973$ says survivors are not integration artifacts; the cross-platform $\rho=0.791$ says they reproduce under orthogonal technology. Because the Sanger CERES panel overlaps discovery lines almost completely, even the $\rho=0.791$ is cross-platform, not out-of-cohort — and we present it as the honest headline precisely so the result is not over-read. This is the same discipline the anchor’s PARP1 vignette encodes: report the weaker true statement.

Limitations (stated plainly). (1) *Not clinically proven.* All findings are computational/preclinical; CRISPR-derived SL cannot be clinically validated here, which requires prospective trial data. No axis is labeled clinically proven; the top rung is a reproducible, clinically actionable node with trial precedent. (2) *Cross-platform, not out-of-cohort.* The L4 replication shares 315/318 cell lines with discovery; a truly held-out cohort was not testable in the available CERES version. (3) *Modality blind spot.* Gene knockout cannot model drug-trapping mechanisms (e.g., PARP trapping), depressing Layer-1 recall for that modality (1/5) and requiring those axes to be validated pharmacologically, not by dependency alone. (4) *Discovery-only nominations.* The 30 E5 axes are hypotheses, not validated targets, and 46/58 axes have not been literature-searched. (5) *Anchor-specific pharmacology.* The MBD4 ATR result derives from GDSC2

pan-cancer pharmacology with confound controls; context-specific SL can be non- resolving in naive pan-cancer comparisons, and the appropriate confirmatory experiment is isogenic. (6) *Observational dependency data* cannot establish causality on their own; the framework mitigates but does not eliminate residual confounding.

Outlook. The framework is designed to be extended: additional anchors worked to Layer 3, truly out-of-cohort replication when a non-overlapping screen panel becomes available, and prospective validation of the highest-trust axes. Its value is procedural — a reproducible, receipt-backed path from a dependency signal to a graded SL hypothesis that states its own ceiling.

Figures

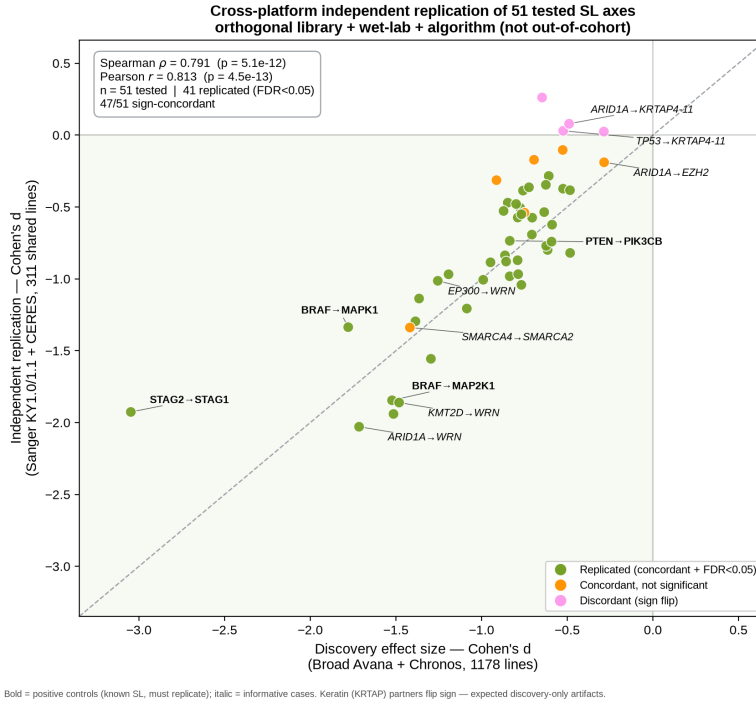


Figure 1: **Cross-platform independent replication (Layer 4).** Paired SL effect sizes for the pan-cancer portfolio on the Broad discovery platform (Avana + Chronos, DepMap 24Q4) versus the orthogonal Sanger Project Score platform (KY1.0/1.1 library, CERES pipeline; Behan 2019). Of 58 axes, 51 were testable and 41 replicated at $FDR < 0.05$, with 47/51 sign-concordant; Spearman $\rho = 0.79$ ($p = 5.1e-12$), Pearson $r = 0.81$ ($n = 51$). Panels overlap in cell lines, so this is cross-platform (orthogonal library, wet-lab, and algorithm) rather than out-of-cohort replication.

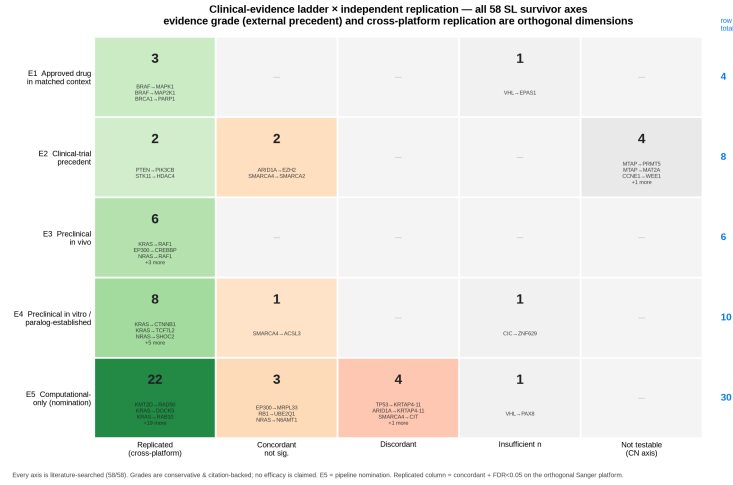


Figure 2: **Clinical evidence ladder (Layer 5)**. Distribution of the 58 driver->partner SL axes across five evidence grades, from E1 (approved drug in a genotype-matched context) to E5 (computational-only nomination). Grades: E1=4, E2=8, E3=6, E4=10, E5=30. The ladder separates a reproducible dependency signal from a clinically graded, actionable hypothesis and makes the computational-only fraction explicit.

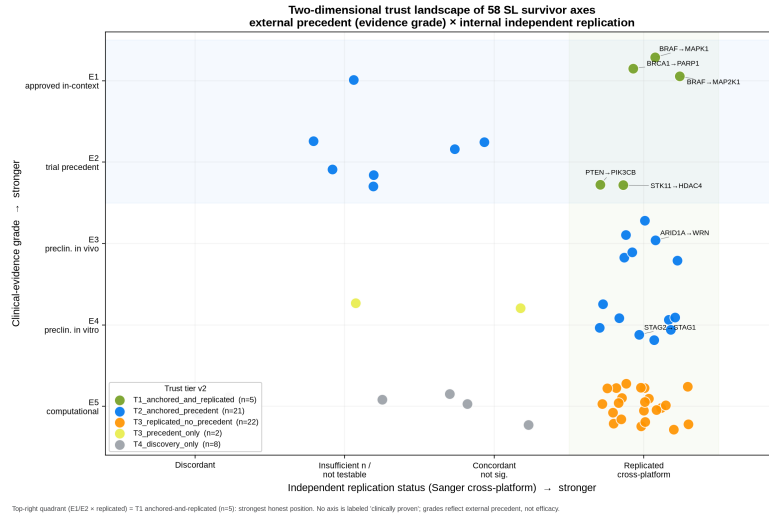


Figure 3: **Two-dimensional trust-tier landscape (Layer 5)**. Each axis is placed by clinical anchoring (precedent) against cross-platform replication status. Five axes are both clinically anchored and cross-platform replicated (T1); discovery-only nominations without precedent occupy the lower-left. This two-dimensional scheme avoids collapsing “reproducible” and “clinically supported” into a single score.

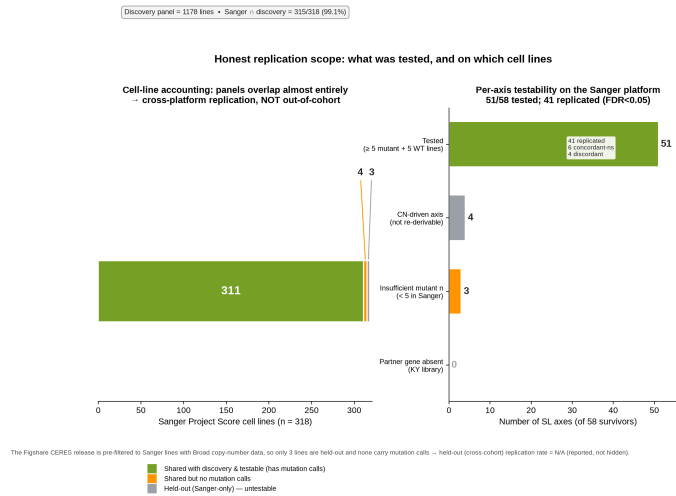


Figure 4: **Replication overlap (Layer 4)**. Concordance between sign agreement and FDR-significant replication across the 51 testable axes, showing the subset (41) that meets both the sign-concordance and FDR<0.05 criteria used to call an axis replicated.

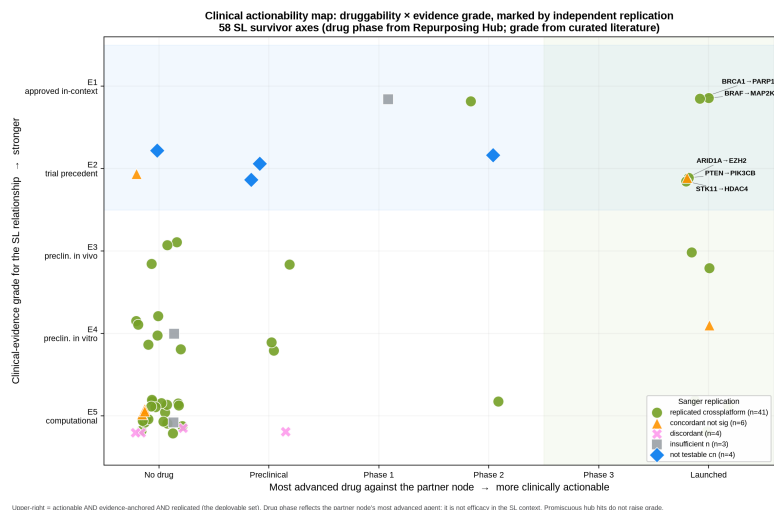


Figure 5: **Actionability map (Layer 5)**. MOA-consistent druggability and clinical precedent across the portfolio: 13/58 axes have an MOA-consistent druggable agent in any phase, 9/58 reach late-phase (Phase 3+/approved), and 17/58 are druggable-or-precedent. Literature was searched for 12 axes; the remaining 46 are unexamined (not “no precedent”).

Methods

Dependency data. Discovery used DepMap 24Q4 integrated Chronos gene effect (CRISPRGeneEffect; 1178 lines x 17,916 genes) (Meyers et al. 2017). Integration-robustness used DepMap 24Q4 per-screen gene effect (ScreenGeneEffect; 1416 screens collapsed to 1186 models). The MBD4 anchor cohort and expression analyses used DepMap 24Q2. Cross-platform replication used Sanger Project Score CERES (Behan et al. 2019; Figshare 10.6084/m9.figshare.9116732; KY1.0/1.1 library (Tzelepis et al. 2016; Behan et al. 2019)).

Genotype calls. Loss-of-function status was called from DepMap OmicsSomaticMutations using the LikelyLoF annotation; missense/passenger variants were purged before stratification. Model metadata (lineage, MSI, subtype) came from DepMap Model.csv.

Differential dependency and pharmacology. SL is a directional hypothesis (driver-altered lines are *more* dependent / *more* drug-sensitive than wild-type), so the discovery engine uses **one-sided** Mann-Whitney tests (`alternative="less"` on gene effect / drug-response) with Benjamini-Hochberg FDR control; the MBD4-anchor GDSC2 drug-response comparisons (LN_IC50, AUC, Z-score) are reported one-sided on the same directional basis (corresponding two-sided p-values are 2x the one-sided value and are noted in-text for the primary ceralasertib result). Effect sizes are reported as Cohen’s

d. Cross-platform concordance is reported as Spearman rho and Pearson r on paired effect sizes; replication at the axis level required sign concordance and $FDR < 0.05$ (Benjamini-Hochberg).

Confound stress tests (anchor). MSI-H exclusion, *TP53*-stratified comparison, leave-one-out robustness, and lineage-restricted comparison, each recomputed independently.

Actionability. Druggability/actionability annotation applied an MOA-consistency filter (inhibitor/antagonist/degrader terms with target-set size ≤ 8) to the Broad Drug Repurposing Hub, plus ClinPGx relationship and drug-label tables; promiscuous-only target annotations were flagged and not counted.

Reproducibility. The validation engine (`sl_clinical_validation.py`, `run_clinical_validation.py`) regenerates the pan-cancer benchmark statistics deterministically; frozen receipts (evidence ladder, benchmark summaries, receipts JSON) accompany this manuscript and back every in-text number (see Data and Code Availability and `SUPPLEMENTAL_DATA_PROVENANCE.md`). The MBD4 PARP1/RNF144A falsification was re-derived directly from the repository mutations table (`OmicSomaticMutations.parquet`; MBD4 LikelyLoF=True yields 21 models, 19 with expression) and the per-line expression receipt (`depmap_expression_axis_genes.csv`, 1,673 lines): LOF-group medians reproduce exactly (PARP1 6.77) and the PARP1-PARPi correlation reproduces (Spearman rho=-0.42, n=488). The p-values reported above use this repository slice (19 LOF vs 1,654 non-LOF comparator); the internal receipt value (p=0.605) was computed against the full DepMap `OmicExpression.csv` (19 LOF vs 1,498 non-LOF), which is not redistributed here. Both comparator pools give the same conclusion (no PARP1 elevation in MBD4-LOF).

Data and Code Availability

Public inputs: DepMap 24Q2/24Q4 (depmap.org); GDSC2 (cancerrxgene.org); Sanger Project Score CERES (Figshare 10.6084/m9.figshare.9116732); Broad Drug Repurposing Hub; ClinPGx. Discovery/validation code and frozen receipts are in the `fjkiani/Synthetic-Lethality` repository (validation engine on branch `feat/pancancer-sl-5layer-validation`, commit 90a56b7; MBD4 anchor artifacts under `00-mbd4-manuscript/mbd4_parp_response/`). The complete pan-cancer discovery and five-layer validation pipeline, including all frozen benchmark summaries and the master SL matrix that back every in-text number, is available at https://github.com/fjkiani/Synthetic-Lethality/tree/main/sl_discovery_pancancer. Supplementary tables (evidence ladder, cross-platform and cross-screen benchmark summaries, actionability) and a claims->receipt map are provided with this manuscript under `sl_platform_manuscript/`.

Competing Interests

The authors are affiliated with CrisPRO.org and Rutgers University. This work is Research Use Only and is not intended to guide diagnosis, prognosis, or treatment selection.

Supplementary Materials

Machine-readable CSVs are archived under [suplimentary/](#) (full column set). Condensed views below use the same receipt-backed values.

Table S1. Evidence ladder and trust-tier assignment (58 axes)

Table S1: Evidence ladder and trust-tier assignment (58 axes). Trust codes: T1 = T1_anchored_and_replicated; T2 = T2_anchored_precedent; T3p = T3_precedent_only; T3r = T3_replicated_no_precedent; T4 = T4_discovery_only. Sanger rep = Y/N at FDR<0.05 on Project Score CERES. Full CSV under [suplimentary/](#).

Driver	Partner	Mode	E	Trust	Sang.	FDR	Cohen <i>d</i>
VHL	EPAS1	LOF	E1	T2	N	—	-2.041
BRAF	MAPK1	ACT	E1	T1	Y	2.6×10^{-5}	-1.336
BRAF	MAP2K1	ACT	E1	T1	Y	2.6×10^{-5}	-1.845
BRCA1	PARP1	LOF	E1	T1	Y	0.008	-0.820
ARID1A	EZH2	LOF	E2	T2	N	0.084	-0.189
SMARCA4	SMARCA2	LOF	E2	T2	N	0.053	-1.339
CCNE1	WEE1	CN_gain	E2	T2	N	—	—
CCNE1	PKMYT1	CN_gain	E2	T2	N	—	—
MTAP	PRMT5	CN_loss	E2	T2	N	—	—
MTAP	MAT2A	CN_loss	E2	T2	N	—	—
PTEN	PIK3CB	LOF	E2	T1	Y	0.002	-0.736
STK11	HDAC4	LOF	E2	T1	Y	0.021	-1.296
ARID1A	WRN	LOF	E3	T2	Y	7.3×10^{-8}	-2.029
EP300	CREBBP	LOF	E3	T2	Y	5.8×10^{-4}	-0.885
EP300	WRN	LOF	E3	T2	Y	0.004	-1.014
KMT2D	WRN	LOF	E3	T2	Y	1.7×10^{-6}	-1.861
KRAS	RAF1	ACT	E3	T2	Y	0.006	-0.507
NRAS	RAF1	ACT	E3	T2	Y	0.001	-0.968
SMARCA4	ACSL3	LOF	E4	T3p	N	0.066	-0.314
CIC	ZNF629	LOF	E4	T3p	N	—	-1.905
ARID1A	ARID1B	LOF	E4	T2	Y	0.001	-1.007
CDKN2A	FOSL1	LOF	E4	T2	Y	1.8×10^{-4}	-1.042
KRAS	CTNNB1	ACT	E4	T2	Y	0.035	-0.470
KRAS	TCF7L2	ACT	E4	T2	Y	0.020	-0.528
NRAS	SHOC2	ACT	E4	T2	Y	2.6×10^{-5}	-1.940
RB1	E2F3	LOF	E4	T2	Y	1.1×10^{-4}	-1.556
RB1	CCNE1	LOF	E4	T2	Y	0.004	-0.799
STAG2	STAG1	LOF	E4	T2	Y	0.045	-1.926
EP300	MRPL33	LOF	E5	T4	N	0.361	-0.103
NRAS	N6AMT1	ACT	E5	T4	N	0.356	-0.172
RB1	UBE2Q1	LOF	E5	T4	N	0.135	-0.540
ARID1A	KRTAP4-11	LOF	E5	T4	N	0.682	0.079
KRAS	STK33	ACT	E5	T4	N	0.505	0.024
SMARCA4	CIT	LOF	E5	T4	N	0.870	0.261
TP53	KRTAP4-11	LOF	E5	T4	N	0.662	0.030
VHL	PAX8	LOF	E5	T4	N	—	-1.936

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Driver	Partner	Mode	E	Trust	Sang.	FDR	Cohen d
ARID1A	MRPL14	LOF	E5	T3r	Y	0.046	-0.346
ARID1A	RPL22L1	LOF	E5	T3r	Y	3.7×10^{-6}	-1.207
BRAF	ELAVL1	ACT	E5	T3r	Y	0.003	-0.982
BRAF	CYB5R4	ACT	E5	T3r	Y	6.0×10^{-4}	-0.968
BRAF	SYNCRIP	ACT	E5	T3r	Y	0.029	-0.575
BRAF	PEA15	ACT	E5	T3r	Y	0.033	-1.137
CDKN2A	EGFR	LOF	E5	T3r	Y	0.013	-0.363
CDKN2A	ITGA3	LOF	E5	T3r	Y	0.043	-0.574
CDKN2A	TUBB4B	LOF	E5	T3r	Y	5.1×10^{-4}	-0.771
EP300	CPOX	LOF	E5	T3r	Y	0.034	-0.386
KMT2D	RAD50	LOF	E5	T3r	Y	1.8×10^{-4}	-0.871
KMT2D	SMARCD1	LOF	E5	T3r	Y	0.004	-0.536
KMT2D	DHX29	LOF	E5	T3r	Y	0.039	-0.373
KRAS	DOCK5	ACT	E5	T3r	Y	5.1×10^{-4}	-0.837
KRAS	RAB10	ACT	E5	T3r	Y	0.038	-0.284
PTEN	MLST8	LOF	E5	T3r	Y	0.032	-0.479
PTEN	RICTOR	LOF	E5	T3r	Y	0.004	-0.692
PTEN	PARS2	LOF	E5	T3r	Y	0.009	-0.623
PTEN	HSD17B12	LOF	E5	T3r	Y	0.036	-0.551
PTEN	BRD2	LOF	E5	T3r	Y	0.032	-0.383
PTEN	MTG1	LOF	E5	T3r	Y	0.004	-0.741
SMARCA4	ATP5PO	LOF	E5	T3r	Y	0.046	-0.880

Table S2. Cross-platform replication (Sanger CERES vs Broad discovery)

Table S2: Cross-platform replication (Sanger CERES vs Broad discovery). Replicated = FDR<0.05 on Sanger. Empty cells = not testable / missing. Full CSV under `supplimentary/`.

Driver	Partner	Mode	n_{mut}	n_{wt}	Sang. d	FDR	Disc. d	Rep.
BRAF	MAPK1	ACT	18	293	-1.336	2.6×10^{-5}	-1.779	Y
BRAF	MAP2K1	ACT	18	293	-1.845	2.6×10^{-5}	-1.523	Y
KRAS	CTNNB1	ACT	65	246	-0.470	0.035	-0.847	Y
KRAS	TCF7L2	ACT	65	246	-0.528	0.020	-0.872	Y
KRAS	RAF1	ACT	65	246	-0.507	0.006	-0.775	Y
EP300	CREBBP	LOF	41	270	-0.885	5.8×10^{-4}	-0.948	Y
NRAS	SHOC2	ACT	13	298	-1.940	2.6×10^{-5}	-1.514	Y
MTAP	PRMT5	CN_loss	0	0	—	—	—	N
RB1	E2F3	LOF	19	292	-1.556	1.1×10^{-4}	-1.296	Y
KMT2D	RAD50	LOF	36	275	-0.871	1.8×10^{-4}	-0.790	Y
KRAS	DOCK5	ACT	65	246	-0.837	5.1×10^{-4}	-0.864	Y
NRAS	RAF1	ACT	13	298	-0.968	0.001	-1.193	Y
KRAS	RAB10	ACT	65	246	-0.284	0.038	-0.608	Y
STAG2	STAG1	LOF	9	302	-1.926	0.045	-3.047	Y
BRAF	ELAVL1	ACT	18	293	-0.982	0.003	-0.836	Y
BRAF	CYB5R4	ACT	18	293	-0.968	6.0×10^{-4}	-0.787	Y
PTEN	MLST8	LOF	27	284	-0.479	0.032	-0.799	Y
SMARCA4	ACSL3	LOF	11	300	-0.314	0.066	-0.913	N
CIC	ZNF629	LOF	4	307	—	—	-1.905	N
KMT2D	SMARCD1	LOF	36	275	-0.536	0.004	-0.635	Y
BRAF	SYNCRIP	ACT	18	293	-0.575	0.029	-0.705	Y
ARID1A	MRPL14	LOF	40	271	-0.346	0.046	-0.625	Y
RB1	CCNE1	LOF	19	292	-0.799	0.004	-0.615	Y
CDKN2A	FOSL1	LOF	27	284	-1.042	1.8×10^{-4}	-0.768	Y
PTEN	RICTOR	LOF	27	284	-0.692	0.004	-0.707	Y
EP300	CPOX	LOF	41	270	-0.386	0.034	-0.758	Y
BRAF	PEA15	ACT	18	293	-1.137	0.033	-1.365	Y

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Driver	Partner	Mode	n_{mut}	n_{wt}	Sang. d	FDR	Disc. d	Rep.
VHL	PAX8	LOF	3	308	—	—	-1.936	N
ARID1A	WRN	LOF	40	271	-2.029	7.3×10^{-8}	-1.715	Y
TP53	KRTAP4-11	LOF	77	234	0.030	0.662	-0.524	N
CDKN2A	EGFR	LOF	27	284	-0.363	0.013	-0.724	Y
KMT2D	DHX29	LOF	36	275	-0.373	0.039	-0.525	Y
PTEN	PIK3CB	LOF	27	284	-0.736	0.002	-0.836	Y
ARID1A	RPL22L1	LOF	40	271	-1.207	3.7×10^{-6}	-1.086	Y
CDKN2A	ITGA3	LOF	27	284	-0.574	0.043	-0.789	Y
EP300	MRPL33	LOF	41	270	-0.103	0.361	-0.527	N
KMT2D	WRN	LOF	36	275	-1.861	1.7×10^{-6}	-1.482	Y
RB1	UBE2Q1	LOF	19	292	-0.540	0.135	-0.750	N
PTEN	PARS2	LOF	27	284	-0.623	0.009	-0.589	Y
CDKN2A	TUBB4B	LOF	27	284	-0.771	5.1×10^{-4}	-0.623	Y
STK11	HDAC4	LOF	9	302	-1.296	0.021	-1.386	Y
NRAS	N6AMT1	ACT	13	298	-0.172	0.356	-0.694	N
EP300	WRN	LOF	41	270	-1.014	0.004	-1.256	Y
ARID1A	ARID1B	LOF	40	271	-1.007	0.001	-0.991	Y
PTEN	HSD17B12	LOF	27	284	-0.551	0.036	-0.768	Y
PTEN	BRD2	LOF	27	284	-0.383	0.032	-0.484	Y
PTEN	MTG1	LOF	27	284	-0.741	0.004	-0.592	Y
ARID1A	KRTAP4-11	LOF	40	271	0.079	0.682	-0.490	N
SMARCA4	ATP5PO	LOF	11	300	-0.880	0.046	-0.857	Y
SMARCA4	CIT	LOF	11	300	0.261	0.870	-0.647	N
MTAP	MAT2A	CN_loss	0	0	—	—	—	N
VHL	EPAS1	LOF	3	308	—	—	-2.041	N
ARID1A	EZH2	LOF	40	271	-0.189	0.084	-0.285	N
SMARCA4	SMARCA2	LOF	11	300	-1.339	0.053	-1.419	N
KRAS	STK33	ACT	65	246	0.024	0.505	-0.287	N
CCNE1	WEE1	CN_gain	0	0	—	—	—	N
CCNE1	PKMYT1	CN_gain	0	0	—	—	—	N
BRCA1	PARP1	LOF	9	302	-0.820	0.008	-0.484	Y

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